Thm. Let f be an analytic in the disk $|z-z_0|<R$, and $\{z_n\}$ be a sequence of distinct points converging to z_0. If $f(z_n)=0$ for all $n\in\mathbb{N}$, then $f\equiv 0$ in $|z-z_0|<R$.

Proof. Since f is analytic, the taylor expansion

$$f(z)=a_0+\sum_{k=1}^{\infty}a_k(z-z_0)^k$$

is valid, in $|z-z_0|<R$. Since f is continuous,

$$\lim_{n\to\infty}f(z_n)=f(z_0)=a_0.$$

so $a_0=0$. So, write

$$f(z)=(z-z_0)\cdot\left(a_1+\sum_{k=2}^{\infty}a_k(z-z_0)^{k-1}\right)$$

let $z=z_n$, so

$$\lim_{n\to\infty}\frac{f(z_n)}{z_n-z_0}=a_1=0,$$ inductively, $a_0=a_1=\cdots=0$.

Corollary. If f is analytic at z_0, then there is $r>0$ such that

$$f(z)\neq 0 \text{ for all } 0<|z-z_0|\leq r$$

or $f\equiv 0$

Proof. If no such $r>0$ exists, then for each $n\in\mathbb{N}$, there is $z_n\in B_{\frac{1}{n}}(z_0)$ s.t.

$$f(z_n)=0.$$

Clearly $z_n\to z_0$, so $f\equiv 0$.

「The Uniqueness Theorem.

Let f be an analytic in a domain D, and that $\{z_n\}$ is a sequence of distinct points, converging $z_0 \in D$.

If $f(z_n)=0$ for all $n \in \mathbb{N}$, then $f \equiv 0$ throughout D.

Proof. Define two sets:

$$A := \{a \in D \mid f \equiv 0 \text{ on } B_r(a) \text{ for some } r>0\}$$

$$B := \{a \in D \mid f \neq 0 \text{ on } B_r(a) \setminus \{a\} \text{ for some } r>0\}.$$

⌈Lemma. Let f be an analytic in the $B_R(z_0)$.
If the set $Z(f) = f^{-1}[\{0\}]$ has a limit point in $B_R(z_0)$,
then $f \equiv 0$ on $B_R(z_0)$.

Proof. Let $z_0 \in B_R(z_0)$ be the limit point of $Z(f)$, by the hyphothesis.
Since f is analytic, the representation

$$f(z) = a_0 + \sum_{k=1}^{\infty} a_k (z - z_0)^k$$

is valid, and since z_0 is the limit point of $Z(f)$, there is a sequence $\{z_n\}$ in $Z(f)$, converging to z_0. Therefore,

$$\lim_{n \to \infty} f(z_n) = f(z_0) = a_0$$

is zero. Then, now,

$$f(z) = \sum_{k=1}^{\infty} a_k (z - z_0)^k = (z - z_0) \cdot \left(a_1 + \sum_{k=2}^{\infty} a_k (z - z_0)^{k-1} \right)$$

And, similarly, we can construct $\{z_n\}$ not containing z_0, so that

$$\frac{f(z_n)}{z_n - z_0} \to 0$$

, so a_1 is also 0. Inductively, we obtain $a_n = 0$ for all $n \in \mathbb{N}$. ☐

⌈Corollary. If f is analytic at z_0, then either
$f \equiv 0$ in some $B_r(z_0)$ or $f \not\equiv 0$ in some $B_{r'}(z_0) \setminus \{z_0\}$.

Proof. If there is no such r' exists, that is, for all $r' > 0$, there is $z \in B_{r'}(z_0)$ s.t. $f(z) = 0$,
then $f \equiv 0$ for some neibor, by the Lemma above.

⌈Uniqueness Theorem.
Let f be an analytic in a domain D, and $Z(f)$ has a limit point in D.
Then $f \equiv 0$ on D.